CHEM 120 MT #2 2022 (1) - Practice midterm for chem!

General Chemistry 2 (McGill University)

Q1 (REPORT ALL VALUES WITH 4 SIGNIFICANT FIGURES) 10 Points

Consider the following reactions and associated equilibrium constants:

$$A(g) \rightleftharpoons C(g) \qquad K_c = 1.700 * 10^4$$

$$D(g) \rightleftharpoons 3B(g) \qquad K_c = 1.000 * 10^{-9}$$

a. Use information above to calculate K_c for the overall reaction

$$2A(g) + 3B(g) \rightleftharpoons 2C(g) + D(g)$$

b. Calculate the equilibrium concentrations of A, B, C, D starting with 0.2000 mol A and 0.5000 mol of B combined in total volume of 1.00 L. Note that none of the concentrations can be zero.

Q2 (REPORT ALL VALUES WITH 4 SIGNIFICANT FIGURES) 10 Points

Consider the following chemical process:

$$AgClNH_3(s) \rightleftharpoons AgCl(s) + NH_3(g)$$

The equilibrium constant is found to be $K_p = 0.0560$ (with gas pressures measured in bar) at 16.3 C. It is an endothermic reaction as written.

a. Describe what chemical changes would occur if you prepare a system of ammonia gas (NH_3) in a chamber containing plenty of solid AgCl and $AgClNH_3$ with an initial pressure ammonia at (0.2560 bar) at 16.3 C.
b. Once the system reaches equilibrium, describe a way to change the volume of the container to reduce the number of moles of NH_3 present.
c. When the system is immersed in an ice bath lowering the temperature to 0 C, the pressure of Ammonia at equilibrium (with all three species still present) was found to be 0.0227 bar. What is the molar enthalpy of the reaction?

Q3 (REPORT ALL VALUES WITH 4 SIGNIFICANT FIGURES) 10 points

Consider the following reaction carried out a 1000 K:

$$A\ (g) \rightleftharpoons B\ (g) + C_2\ (g) \qquad K_p = 0.1000$$

a. At 1000 K the reaction is prepared by adding A until the pressure is 1 bar, with no B or C_2 present initially. The reactor is sealed and the reaction is allowed to come to equilibrium. What is the partial pressure of each species present once the reaction comes to equilibrium?

b. Using Le Chatelier's principle describe the changes in the amounts of each species present if the size of the reactor is increased after the system has reached equilibrium?

c. When the reaction is carried out at 800 K from the same starting condition you observe substantially lower amounts of B and C_2 formed once equilibrium is reached. Predict the sign of ΔH^0 via Le Chatelier's principle.

Q4 (REPORT ALL VALUES TO TWO DECIMAL PLACES) 10 Points

The following reaction is known to have **a very large** equilibrium constant.

$$4\,NH_3(g) + 5\,O_2(g) \rightleftharpoons 4\,NO(g) + 6\,H_2O(liq)$$

a . Predict the sign of ΔS^0_{rxn} , with a brief explanation.

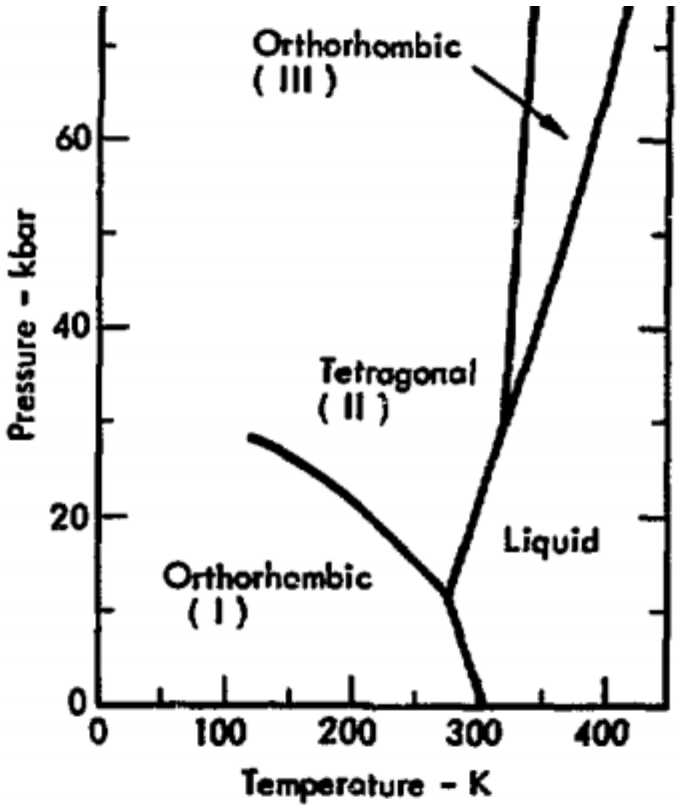
b. Based on your answer to part a. predict the sign of ΔH^0_{rxn}

c. Using the tabulated values from Appendix G the OpenStax textbook find

ΔG^0_{rxn} , ΔH^0_{rxn} , ΔS^0_{rxn} for the reaction.

question removed

Q6 5 Points



The figure above shows the phase diagram of the element Gallium, showing three possible solid phases (orthorhombic (I), tetragonal (II), and orthorhombic (III)) and the liquid phase.

a. If a measurement showed solid gallium deposits deep inside of meteorite found in the orthorhombic (III) phase, based on the diagram here, what is lowest pressure possible?

b. Gallium is discovered on a new planet, and preliminary measurements show that there are oceans of gallium with gallium-bergs ***floating*** in them (like an iceberg, but solid gallium). What is the approximate temperature range possible for this planet (making no assumption of the pressure)?

Q7 (REPORT ALL VALUES WITH 4 SIGNIFICANT FIGURES) 8 points

A new element has been discovered here- McGillium (McG)! The normal boiling point of McGillium is 1160 K. It has one solid phase (denoted as McG(i)), which melts at 612 K. The solid form floats on top of the liquid.

Unfortunately, you can't measure the enthalpy of vaporization. You are able to measure that the enthalpy of solidification is -9.90 kJ/mol and the enthalpy of deposition is -40.70 kJ/mol.

You also measure the follow heat capacities:

Phase	Heat Capacity (J/(mol*C))
McG(i)	35.5
McG(*liq*)	44.7
McG(*g*)	25.1

Calculate the amount of heat necessary to heat up 2 mol of McGillium from 500 K to 2000 K under normal atmospheric pressure. Report your answer in kilojoules.

Q8 REPORT ALL VALUES WITH 4 SIGNIFICANT FIGURES) 12 Points

10 grams of glucose ($C_6H_{12}O_6$) is dissolved in 90 grams of water. The density of the solution is 1.038 grams/mL.

a. What is the molality and molarity of the resulting solution? The M.W. of glucose is 180.16 grams/mol, the M.W. of water is 18 grams/mol.

b. What is the boiling point of such a solution (K_b = 1.86 C* kg/mol)?

c. What would be the osmotic pressure measured for this solution at 25 C (report the answer in bar)?

d.. To get the same boiling point as in part b. for a solution of $CaCl_2$, how many moles of $CaCl_2$ should be added to 1 kg of water (assume the ideal value for the van't Hoff factor)?